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(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU, RTMNU & MSBTE Mumbai

Department of Computer Science & Engineering

"A Place to Learn, A Chance to Grow"

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Progress Seminar No.-04

Secure Blockchain-Based System for Digital Asset Management and Inheritance

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INDEX



- Introduction
- Problem definition
- Literature Review
- Objectives of the study
- Limitations
- Scope
- Methodology
- Flow Chart
- Implementation
- Tools & Techniques
- Conclusions (expected)
- Publication if any
- Patents/Copyright if any
- References

Introduction

- Existing platforms lack transparent, automated enforcement using blockchain and smart contracts.
- Most solutions do not provide real-time monitoring or AI-driven fraud detection.
- Traditional estate planning cannot handle complex digital assets across multiple platforms.
- Strong security with multi-factor verification, adaptive monitoring, and programmable policies is essential



Problem Definition

The rapid growth of digital assets such as cryptocurrency, online banking credentials, cloud documents, social media accounts, digital contracts, and intellectual property has created major challenges in secure asset management and inheritance. Existing inheritance and estate planning systems are mostly manual, centralized, and unable to efficiently handle digital assets distributed across multiple online platforms. These systems often lack transparency, automation, strong authentication, and secure verification mechanisms.

Current solutions also fail to provide reliable real-time monitoring and AI-driven fraud detection, increasing the risk of identity theft, unauthorized access, false inheritance claims, and data tampering. Traditional methods cannot accurately verify critical events such as death or incapacitation, resulting in legal disputes, delays, and security vulnerabilities during asset transfer processes.

To address these issues, there is a need for a secure, transparent, and automated digital inheritance framework that leverages blockchain technology, smart contracts, encryption, multi-factor authentication, biometrics, and AI-based anomaly detection. The proposed system aims to ensure tamper-proof asset storage, automated inheritance execution, secure nominee verification, and legally compliant digital asset transfer while maintaining privacy, trust, and scalability.

Literature Review

Sr. No.	Paper Title	Author	Publication & Year	Key Points
1	Trusted Wills for Digital Assets with Blockchain	Author not specified	SciDirect, 2025	Emphasizes the critical role of integrating legal frameworks, policy compliance, and cutting-edge digital security technologies to ensure effective digital asset inheritance.
2	The Impact of Digitalization in Inheritance Law	A. A. Akramov	Akramov Qubahan Academic, 2024	Recommends the use of blockchain technology and artificial intelligence to modernize inheritance law and improve efficiency, transparency, and legal certainty.
3	The Law Applicable to Transfer of Digital Assets	B. Yüksel	SSRN, 2023	Explains legal complexities, jurisdictional challenges, and regulatory compliance issues involved in the transfer of digital assets across different legal systems.

Sr. No.	Paper Title	Author	Publication & Year	Key Points
5	Bitcoin: A Peer-to-Peer Electronic Cash System	Nakamoto, S.	2008	Introduced blockchain technology for transparent and tamper-proof record keeping; adaptable for verifying posthumous data transfers.
6	Decentralizing Privacy: Using Blockchain to Protect Personal Data	Zyskind, G., Nathan, O.	IEEE Security & Privacy Workshops, 2015	Discusses privacy-preserving data control methods for secure management of user content.
7	How to Share a Secret	Shamir, A.	Communications of the ACM, 1979	Proposes secret-sharing technique for splitting sensitive data (like passwords or keys) among trusted nominees.
8	Web Security Practices in Sensitive Information Management Systems	Kumar, V., & Saini, A.	Int. Journal of Web Applications, 2021	Describes HTTPS, input sanitization, and encryption for securing sensitive web content.

Sr. No.	Paper Title	Author	Publication & Year	Key Points
9	Toward Secure and Dependable Storage Services in Cloud Computing	Wang, C., Ren, K., Lou, W	IEEE Transactions on Services Computing, 2010	Explores methods for dependable cloud storage and secure document transfer.
10	Digital Estate Planning: Practical Considerations for Security and Access	Furnell, S.	Journal of Information Privacy and Security, 2007	Examines ethical and security aspects of managing digital assets after death.
11	Blockchain: Blueprint for a New Economy	Swan, M.	O'Reilly Media, 2015	Provides use cases of blockchain for secure identity and transparent transactions.
12	The Transport Layer Security (TLS) Protocol Version 1.3	Rescorla, E.	RFC 8446, 2018	Defines encryption standards for secure web communications.

Sr. No.	Paper Title	Author	Publication & Year	Key Points
13	Design and Implementation of Secure User Authentication for Web Applications	Kumar, R., & Verma, S.	Journal of Web Engineering, 2020	Discusses multi-factor authentication for verifying user and nominee identities.
14	The Double Ratchet Algorithm	Marlinspike, M., & Perrin, T.	Signal Protocol Specification, 2016	Describes secure end-to-end encryption used for confidential messaging.
15	Smart Contracts for Automated Asset Transfer	Liu, Y., & Zhang, Q.	IEEE Access, 2020	Explains automation of asset transfer using predefined digital conditions.
16	General Data Protection Regulation (GDPR)	European Parliament	2016	Establishes principles of user consent and data privacy in digital systems.

Objectives of the Study



Multi-factor onboarding with ID, biometrics & social proof.



AI-powered real-time anomaly detection & alerts.



Quantum-resistant encryption for secure storage & transfer



Automated workflows mapped to legal and regulatory frameworks

Limitations

- Absence of universal legal and regulatory frameworks for digital inheritance across different countries.
- Blockchain, AI, and biometric technologies are still evolving and may face accuracy or scalability issues.
- System implementation and maintenance require high technical expertise, infrastructure, and operational cost.



Regulation



Security

Scope

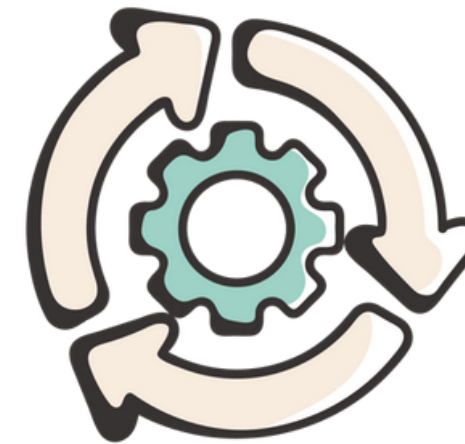
- Provides secure management and storage of digital, financial, legal, and cryptocurrency assets in a single platform.
- Enables automated and transparent inheritance processes using blockchain technology and smart contracts.
- Enhances system security through AI-based fraud detection, biometric verification, and multi-factor authentication.

Methodology

- User-centered design → We keep users at the center, take their feedback, and improve the project in small steps.
Technical sprints → We worked in short phases, each focusing on specific tasks
- like onboarding, monitoring, encryption, alerts, and legal rules.

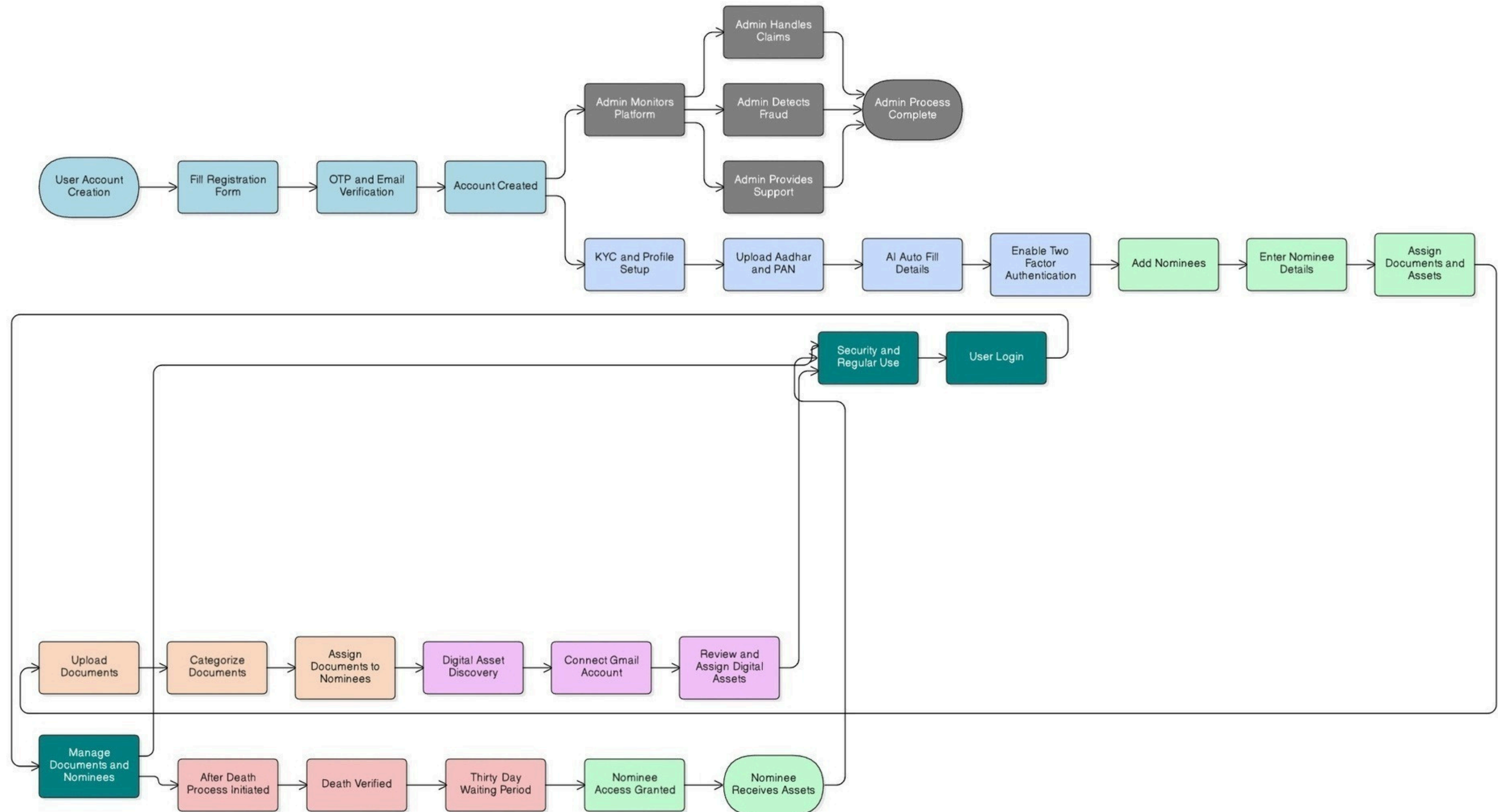


Feedback-driven,
iterative rollouts

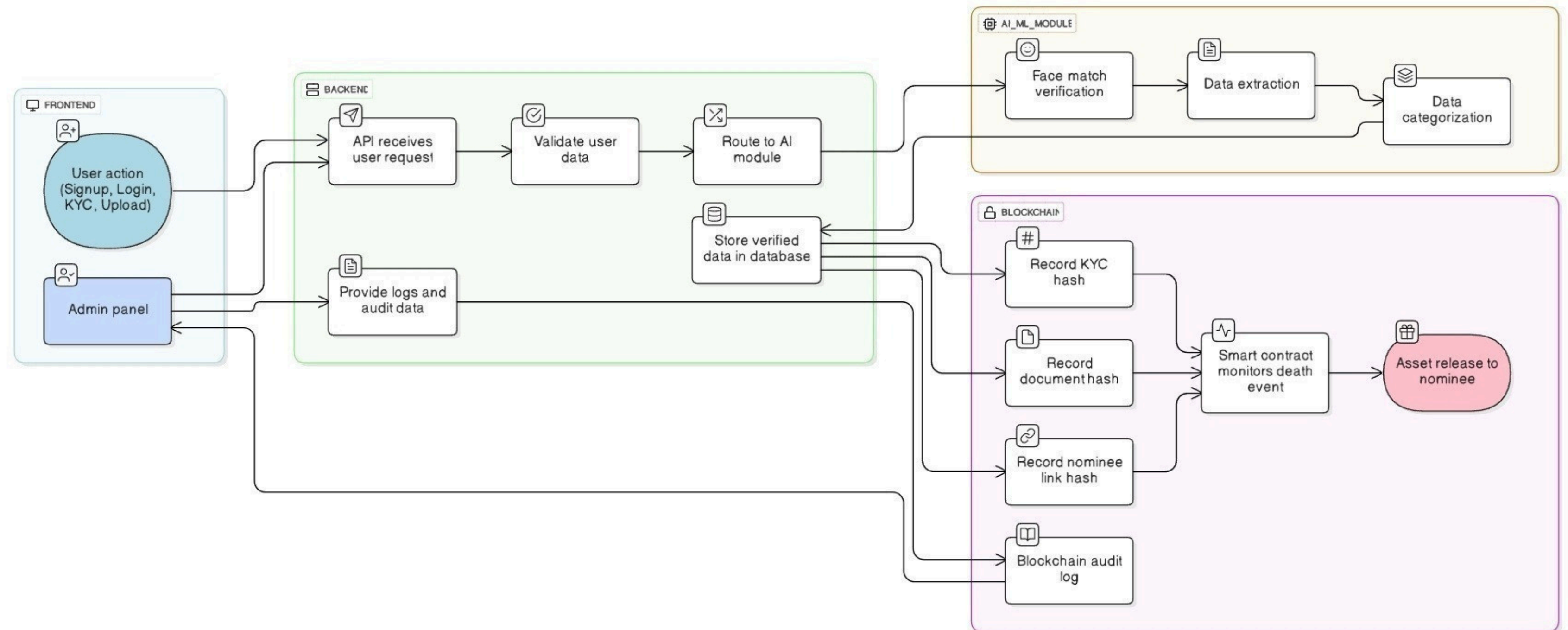


Onboarding, monitoring,
encryption, compliance

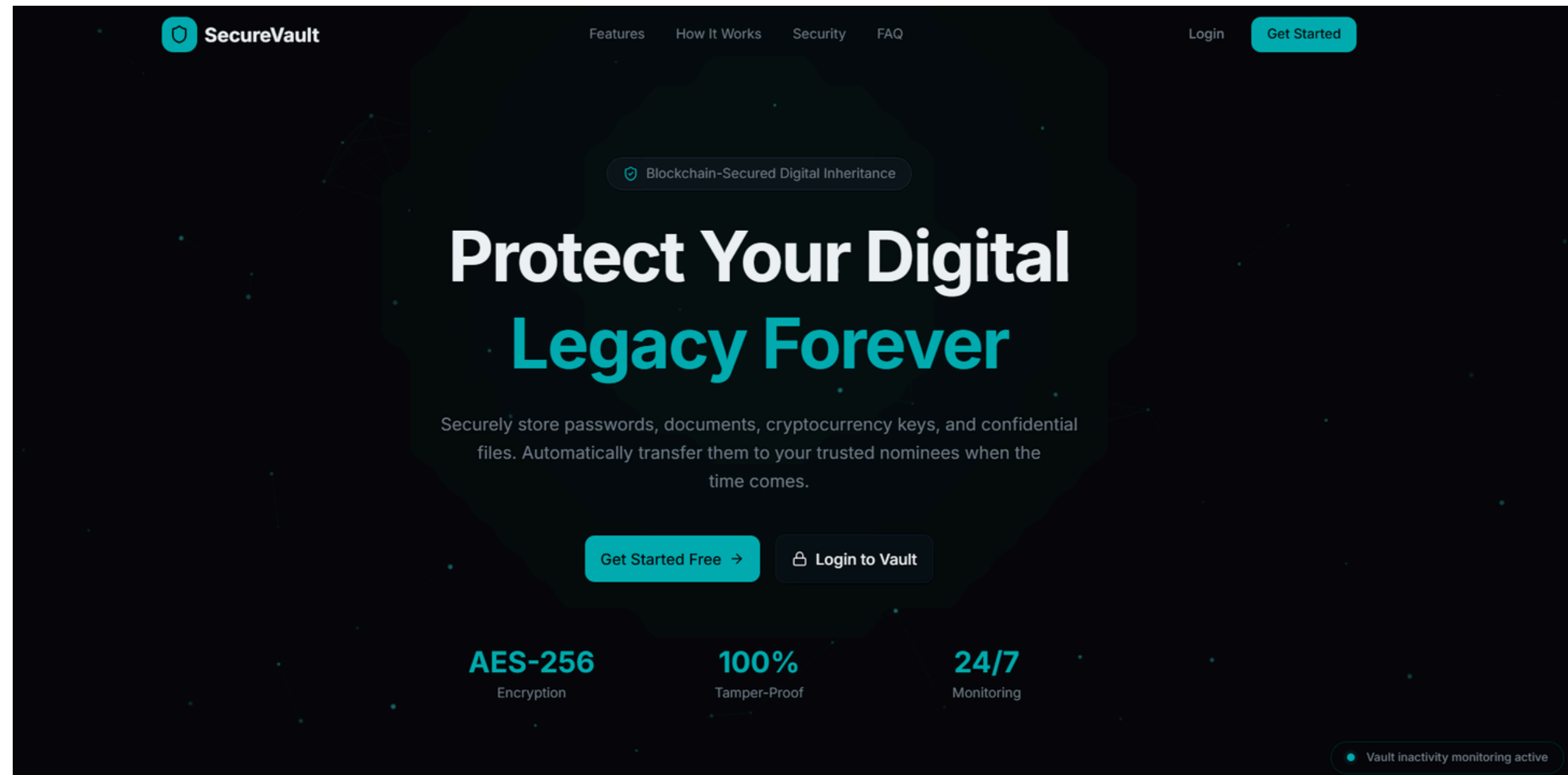
FlowChart



FlowChart



Implementation



This homepage is designed for the SecureVault platform, which provides secure digital asset management and inheritance services using blockchain technology. The interface allows users to safely store passwords, confidential documents, and cryptocurrency keys with advanced AES-256 encryption and multi-factor security. It also supports automated inheritance transfer to trusted nominees through smart contract-based verification. The modern dashboard ensures secure access, real-time monitoring, and tamper-proof protection of sensitive digital assets.

Implementation

← Back to Home



Welcome Back

Login to access your secure vault

Email

you@example.com

Password [Forgot Password?](#)

Enter your password

Login

or

 Continue with Google

Don't have an account? [Sign Up](#)



Create Your Vault

Secure your digital legacy in minutes

 Sign up with Google

OR

Full Name

John Doe

Phone

+91 9876543210

Date of Birth

dd-mm-yyyy

Email

you@example.com

Password

Create a strong password

Confirm Password

Rewrite your password

Continue →



Inactivity Period

Set how long before your assets are transferred to nominees

If you remain inactive (no login) for the selected period, the system will initiate the secure asset transfer process to your designated nominees.

3 Months 6 Months 12 Months 24 Months



Two-Factor Authentication

Add an extra layer of security to your account

Authenticator App

Use an app like Google Authenticator or Authy to generate secure codes.

Enabled Disabled

Enable 2FA



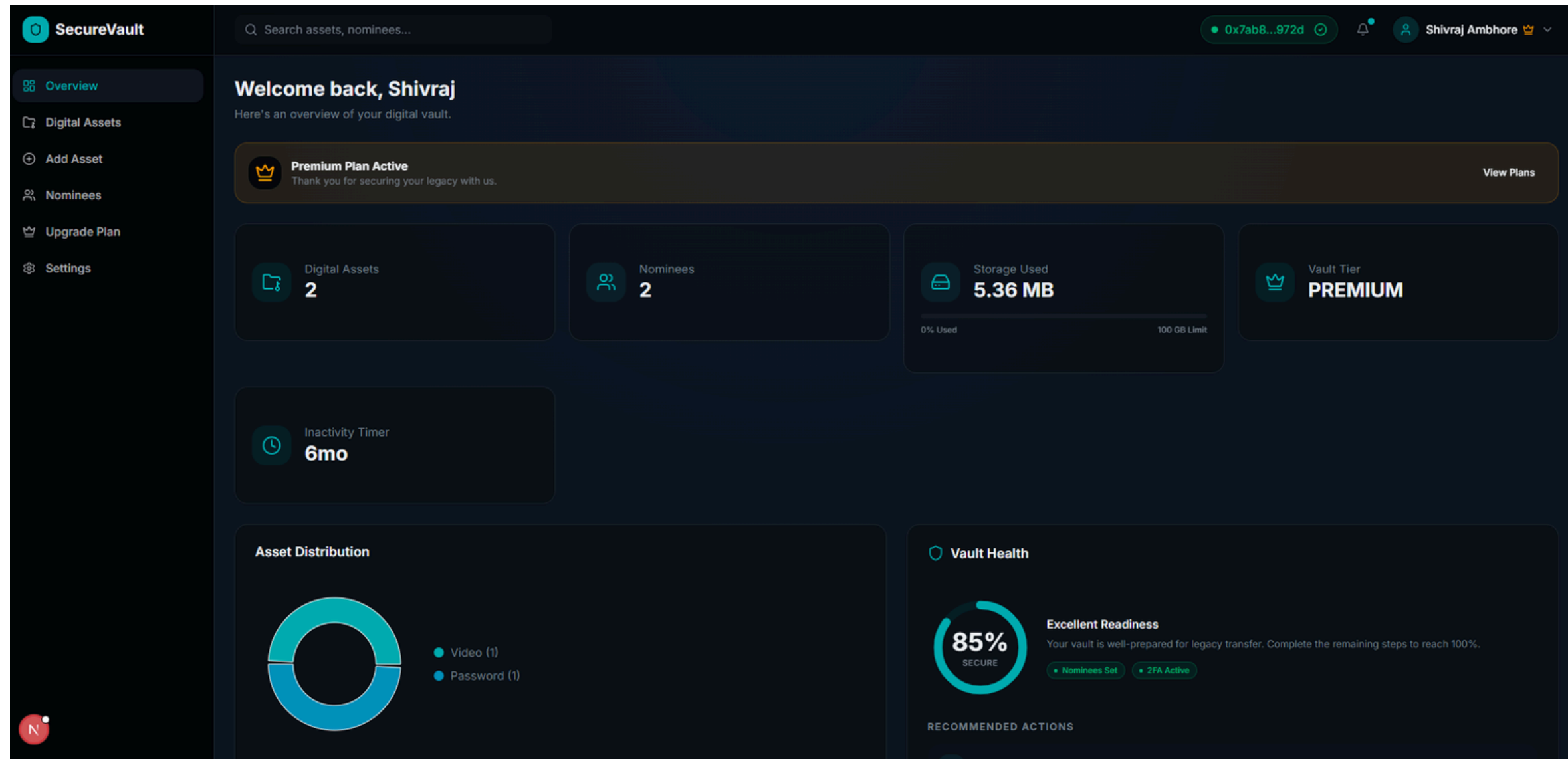
Security Overview

Your vault is protected with AES-256 encryption, blockchain verification, and smart contract automation. All actions are logged immutably on-chain.

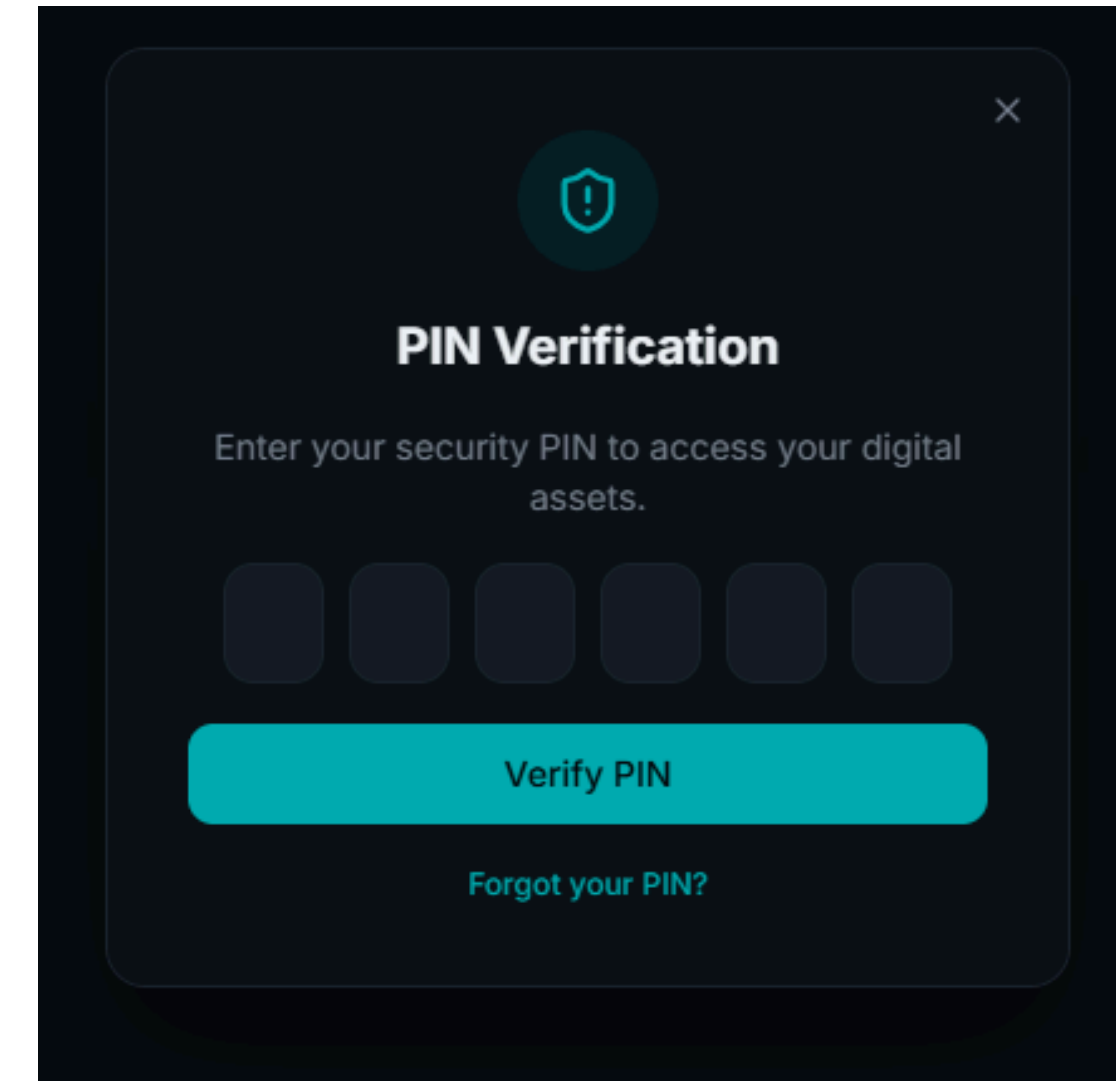
AES-256 JWT Auth Blockchain Smart Contracts

The login and signup pages provide secure user authentication and account creation features, allowing users to safely access and manage their digital assets, nominee details, and inheritance information within the SecureVault platform.

Implementation



This dashboard provides users with a secure overview of their digital vault, including stored assets, nominees, storage usage, and security status. It enables easy management of digital inheritance with real-time monitoring, inactivity tracking, and encrypted asset protection through a user-friendly interface.



This PIN verification page adds an extra layer of security by requiring users to enter a secure PIN before accessing sensitive digital assets.



Implementation

Add Digital Asset

Securely encrypt and store documents, media, or credentials.

Asset Configuration

Asset Category

Image

Video

Document

Text Note

Password

Legal File

Asset Name

Give your asset a clear name

Description (Optional)

Add some context or tags...

Assign Beneficiary (Nominee)

Who should inherit this? ▾

Store in Secure Vault

SecureVault

Search assets, nominees...

0x7ab8...972d

Shivraj Ambhore

Overview

Digital Assets

Add Asset

Nominees

Upgrade Plan

Settings

Digital Vault

Manage and inherit your secure digital legacy.

Search your vault...

All Categories ▾

VIDEO

secuvideo
No description provided
Inheritor: santosh ambhoree 5/3/2026

PASSWORD

om password
No description provided
Inheritor: om 5/3/2026

Nominees

Manage your trusted nominees who will receive your digital assets.

SA

Parent

santosh ambhoree
shivrajgdgjdcoem@gmail.com
+918380870615
1 asset assigned

Edit Remove

O

Child

om
shivrajambhore1111@gmail.com
+918380870615
1 asset assigned

Edit Remove

Implementation

Upgrade Your Vault

Secure more legacy with our premium storage plans.

Free

\$0 one-time

500 MB Storage

- ✓ 500 MB Secure Storage
- ✓ Basic Asset Upload
- ✓ 50MB File Size Limit
- ✓ Standard Support

Upgrade to Free

Pro

\$29 one-time

10 GB Storage

- ✓ 10 GB Secure Storage
- ✓ 500MB File Size Limit
- ✓ Pro Badge on Profile
- ✓ Priority Support
- ✓ Ad-free Experience

Upgrade to Pro

Premium

\$99 one-time

100 GB Storage

- ✓ 100 GB Secure Storage
- ✓ Unlimited File Size
- ✓ Crown Badge 👑
- ✓ Premium Exclusive Theme
- ✓ Advanced Filtering
- ✓ Early Feature Access

Current Plan

Settings

Manage your profile, security, and inheritance settings.

Profile Settings

Update your personal information

Full Name

Shivraj Ambhore

Email

shivrajambhore01@gmail.com

Phone

+918010170071

Save Changes

Change Password

Update your account password

Current Password

Enter current password

New Password

Enter new password

Confirm New Password

Confirm new password

Change Password

The Premium and Settings pages allow users to upgrade their vault services, manage account preferences, configure security options, and customize privacy settings for better protection and control of digital assets.

Tools

Tech Stack (Frontend + Backend)

- React.js / Next.js, Tailwind CSS
- Node.js + Express.js, REST APIs
- MongoDB

Security Framework

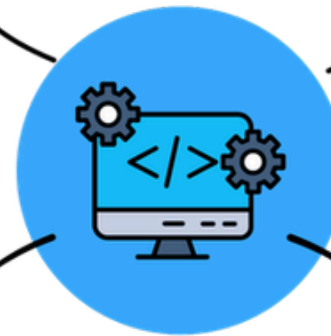
- JWT, 2FA
- RBAC, Input Validation, Rate Limiting, Captcha
- AES-256

AI & Verification

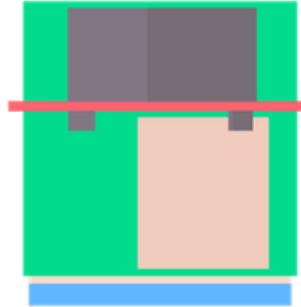
- OCR for certificate validation
- AI/ML anomaly & fraud detection
- API integrations (Hospitals/Govt.) for verification

Deployment & Infrastructure

- Vercel / Netlify (frontend hosting)
- AWS / Heroku / Render (backend + cloud)



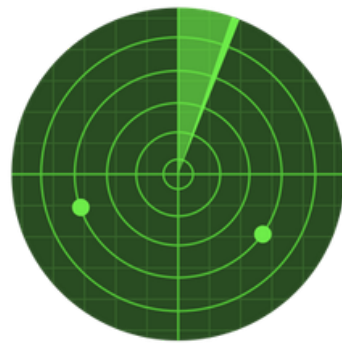
Techniques



"Facial biometrics for multi-layer secure onboarding



Biometric-derived keys with hierarchical RBAC to ensure privacy and access control.



AI-driven anomaly detection with instant fraud alerts and quick response.



Immutable audit trails and multi-party consensus to secure asset transfers.

Conclusions

- **Secure Digital Asset Management** : The proposed system provides a secure and reliable platform for storing, managing, and protecting digital assets such as cryptocurrencies, legal documents, financial records, and online accounts through advanced encryption and blockchain technology.
- **Automated and Transparent Inheritance** : By using blockchain and smart contracts, the system automates the inheritance process, ensuring transparent, tamper-proof, and trustworthy transfer of assets to verified nominees without manual intervention.
- **Advanced Security and Fraud Prevention** : The integration of biometric authentication, multi-factor verification, and AI-based anomaly detection enhances security, minimizes fraud risks, and prevents unauthorized access or false inheritance claims.

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Hackathon Certificate



L & T Certificate



L & T Certificate



Thank You!!!